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Effects of CO_2 enrichment on whole-plant carbon budget of seedlings of *Fagus grandifolia* and *Acer saccharum* in low irradiance

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Abstract Carbon exchange rates (CER) and whole-plant carbon balances of beech (*Fagus grandifolia*) and sugar maple (*Acer saccharum*) were compared for seedlings grown under low irradiance to determine the effects of atmospheric CO_2 enrichment on shade-tolerant seedlings of co-dominant species. Under contemporary atmospheric CO_2, photosynthetic rate per unit mass of beech was lower than for sugar maple, and atmospheric CO_2 enrichment enhanced photosynthesis for beech only. Aboveground respiration per unit mass decreased with CO_2 enrichment for both species while root respiration per unit mass decreased for sugar maple only. Under contemporary atmospheric CO_2, beech had lower C uptake per plant than sugar maple, while C losses per plant to nocturnal aboveground and root respiration were similar for both species. Under elevated CO_2, C uptake per plant was similar for both species, indicating a significant relative increase in whole-seedling CER with CO_2 enrichment for beech but not for sugar maple. Total C loss per plant to aboveground respiration was decreased for beech only because increase in sugar maple leaf mass counterbalanced a reduction in respiration rates. Carbon loss to root respiration per plant was not changed by CO_2 enrichment for either species. However, changes in maintenance respiration cost and nitrogen level suggest changes in tissue composition with elevated CO_2. Beech had a greater net daily C gain with CO_2 enrichment than did sugar maple in contrast to a lower one under contemporary CO_2. Elevated CO_2 preferentially enhances the net C balance of beech by increasing photosynthesis and reducing respiration cost. In all cases, the greatest C lost was by roots, indicating the importance of belowground biomass in net C gain. Relative growth rate estimated from biomass accumulation was not affected by CO_2 enrichment for either species possibly because of slow growth under low light. This study indicates the importance of direct effects of CO_2 enrichment when predicting potential change in species distribution with global climate change.

Key words Shade tolerance · Whole-seedling CER Root respiration · Maintenance respiration CO_2 enrichment

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Introduction

Global atmospheric CO_2 concentration has been increasing from a pre-industrial level of ca. 270 $\mu l\ l^{-1}$ CO_2 in 1870 (Neftel et al. 1985) to a mean value of 354 $\mu l\ l^{-1}$ in 1990 (Keeling and Whorf 1991). Doubling of the pre-industrial level of atmospheric CO_2 is expected by the mid-21st century if present fossil fuel consumption rates continue. In addition to its potential warming effects on climate, elevated atmospheric CO_2 directly affects vegetation by enhancing gas exchange properties and growth (Strain and Cure 1985; Eamus and Jarvis 1989; Bazzaz 1990; Strain 1992). The magnitude of these enhancements by CO_2 enrichment varies with species and with other limiting environmental conditions. Carbon dioxide enrichment often ameliorates stressful conditions such as drought, nutrient deficiencies, and low irradiance (reviewed in Eamus and Jarvis 1989; Bazzaz 1990).

In the forest understory, low irradiance limits growth and eventually survival of tree seedlings. Seedling survival is critical for canopy regeneration in old-growth forests, and positive interaction between low irradiance and elevated CO_2 may favor seedling growth on the forest floor. Carbon dioxide enrichment at low irradiance enhances gas exchange and growth of shade-intolerant seedlings of *Liquidambar* (Tolley and Strain 1984, 1985). Growth stimulation requires an increase in net carbon exchange rates (CER) and may be caused solely by enhanced photosynthesis, by reduced respiration, or a combination of both.

Existing evidence for effects of CO_2 enrichment on dark respiration indicate species-specific responses. Gifford et al. (1985) found either a suppression or no change in root and shoot respiration with atmospheric CO_2 enrichment for five crop species examined. Hrubec et al. (1985) showed higher shoot respiration only during leaf expansion at elevated CO_2, whereas Bunce (1992) has reported a decrease in whole-plant respiration for seedlings of deciduous tree species. Amthor (1991) proposed possible responses of respiration to elevated CO_2, ranging from greater respiration because of increased substrate availability to lower respiration because of changes in tissue composition. If respiration is suppressed by CO_2 enrichment, the magnitude of photosynthetic enhancement should lead to greater seedling longevity and survival in the understory. On the other hand, enhanced photosynthesis may not affect seedling survival if dark respiration increases with CO_2 enrichment. Patterns of daily carbon gain resulting from differential responses of respiration as well as photosynthesis may be greatly changed by elevated CO_2.

Beech (*Fagus grandifolia* Ehrh.)[1] and sugar maple (*Acer saccharum* Marsh.) are the dominant species of many northeastern deciduous forests (Whittaker et al. 1974; Hicks and Chabot 1985). Both species are very shade-tolerant (Fowells 1965), although beech saplings survive in the dense shade of the canopy longer than sugar maple (Poulson and Platt 1989). In the understory, the differential growth of beech and sugar maple saplings has been correlated with high efficiency of leaf display for beech in closed canopy in contrast to broad morphological plasticity of leaf display for sugar maple in light gaps (Canham 1988). For shade-tolerant species, Canham (1989) proposed a gradient of responses to pulses of light through the canopy based on trade-offs between morphological and physiological plasticity. Physiological studies to determine the carbon budgets of 4-year-old seedlings of beech and sugar maple suggest two patterns for growth and survival in low light of the understory: (1) low carbon uptake rate coupled with low root respiration rate enabling the maintenance of a large non-photosynthetic biomass, for beech, and (2) high carbon uptake rate associated with high root respiration rate that may allow greater photosynthetic plasticity but also high maintenance cost of belowground biomass, for sugar maple (Reid 1990). We therefore hypothesized that carbon budgets of these two species might be modified differently under elevated CO_2. Such effects might be important for future forest regeneration.

The objective of this study was to determine the effects of CO_2 enrichment on the patterns of carbon exchange and growth for seedlings of two shade-tolerant tree species grown in low irradiance at current and elevated CO_2. Daily carbon exchange rates (CER), net carbon gain, and resulting growth were compared for seedlings of beech and sugar maple under current and projected elevated atmospheric CO_2.

[1] Nomenclature follows Fowells (1965)

Material and methods

Three-year-old dormant seedlings of beech (*Fagus grandifolia* Ehrh.) and sugar maple (*Acer saccharum* Marsh.) were collected in an old-growth beech-maple forest at Mont St-Hilaire, Rouville County, Québec (45°10' N, 73°10' W) in October 1987. The seedlings were transported to the Duke University Phytotron where they were transplanted into a mixture of sand and gravel (1:2) in 2.7-l containers. Each pot was inoculated with ca. 2.5 cm^3 of native Québec soil to establish local mycorrhizae. The plants were kept dormant until the beginning of the experiment.

Dormancy was broken by exposing the plants to 375 μmol quanta m^{-2} s^{-1} photosynthetic photon flux density (PPFD) at 19°C and 24-h photoperiod. Seedling length at the beginning of the experiment was 20 ± 0.7 cm for beech and 16 ± 0.7 cm for sugar maple. Assignment to treatments was made so that variation was the same in each treatment per species. Thirty-six seedlings per species were transferred in two growth chambers at different intervals to allow gas exchange measurements at similar phenological intervals between treatments and species. Bud break and leaf expansion were thus monitored and aging was based on days after leaf expansion. Both species, which produce one set of leaves per year, leafed out in similar time intervals and produced similar leaf area. Seedlings were grown under low light (65 μmol quanta m^{-2} s^{-1}) at current (350 $\mu l\ l^{-1}$) and elevated (650 $\mu l\ l^{-1}$) atmospheric CO_2 in 0.9-m×1.2-m reach-in growth chambers (Kramer et al. 1970). Chamber CO_2 concentrations were maintained with an automated CO_2 control system (Hellmers and Giles 1979). The temperature regime was 19°C day: 15°C night. Photoperiod was 12 h day: 12 h night. The short photoperiod compensated for the lower irradiance level and duration prevalent under closed canopies. Plants received one quarter strength modified Hoaglands' nutrient solution (Downs and Hellmers 1978) each morning and water each afternoon. Nitrogen content in the nutrient solution was non-limiting in both species (Reid 1990). Plants were rotated in chambers every 10 days to minimize chamber-position effects. Conditions in the growth chambers were assumed to be similar except for the atmospheric CO_2 concentration because environmental conditions within chambers were continuously monitored. Potential chamber effects were considered minimum since the random assignment of treatments to growth chambers through the many experiments of this project yielded similar CO_2 effects on seedlings (Reid 1990).

Simultaneous above- and belowground diurnal gas exchanges were measured using an infrared gas analyzer system (ADC 225) and a two-chambered cuvette incorporated into a standard open-type gas exchange system. The aboveground portion of the cuvette was a 14 cm wide × 21 cm long × 11 cm high rectangular cuvette with nickel-plated copper walls and a Plexiglas top, similar to the one described by Riechers (1983). Stem gas exchange was considered to be negligible, so that aboveground gas exchange was due to the leaves only. Measurements of belowground gas exchange were modified from Lawrence and Oechel (1983). The belowground chamber consisted of a sealed Plexiglas cylinder that contained the whole pot. Air entered from the bottom and was allowed to diffuse to an outlet on top of the chamber. Air flow was adjusted to provide rapid gas exchange and sufficient CO_2 enrichment for accurate measurements. Air flow ranged from 0.73 to 0.80 l min^{-1}. Once the plant was in the cuvette, both chambers were continuously flushed with air at constant flow rate and measurements started after steady-state CERs were reached.

Each plant was sampled continuously for 24 h, alternating between the above- and belowground chambers every 15 min. The first measurements began 19 days after leaf expansion. Plants were measured under the same environmental conditions in which they were grown. Sampling spanned a 70-day period in the 350 $\mu l\ l^{-1}$ CO_2 treatment and 80 days in the 650 $\mu l\ l^{-1}$ CO_2 treatment. Irradiance was provided by a 400-W phosphor-coated General Electric Multi-vapor metal halide lamp (Model MV 400 K 1 BUH) and multiple layers of neutral density shade cloth between the light and the cuvette. Soil temperature was maintained at air temperature using refrigerated water circulated through a copper coil around the pot.

Carbon dioxide efflux due to root respiration was corrected for soil respiration by microorganisms in the following manner. After gas exchange measurements, the seedling was gently removed from the pot minimizing disturbance to the soil. The gravel-sand soil enabled recovery of the root system with negligible loss of fine roots. The plant-free pot was measured again for CO_2 gas exchange after the soil had stabilized for 2 days. The delay was necessary to avoid high CO_2 efflux caused by disturbed microorganisms and to reach steady state although microorganisms' activity may then be lower because of unavailability of new root exudates without the intact root system. These plant-free measurements were averaged for each species and treatment, and used to correct the root gas exchange.

To distinguish maintenance respiration from total respiration, seedlings were placed in the dark for 72 h after measurement of diel gas exchanges (McCree and Kresovich 1978). Preliminary measurements of CER at different intervals following the beginning of the dark treatment indicated a constant respiration rate 72 h after darkening. Carbohydrate reserves are depleted in the dark period until no growth is possible, and respiration measured afterwards is for maintenance. After 72 h, seedlings were measured again in the dark. The plant was harvested, minimizing disturbance to the soil as explained above, following completion of its CER measurements.

Biomass was determined from destructive growth measurements following each series of gas exchange. At the end of the experiment, a few seedlings not measured were harvested. Seedlings were separated into leaves and petioles, stem, large roots (> 1 mm diameter), and fine roots (≤ 1 mm). Tissues were air-dried to constant mass in a forced-air oven at 70°C. For nitrogen (N) analysis, the dry leaf and root material was ground using a Wiley mill and then digested by sulfuric acid-hydrogen peroxide digestion (Wolf 1982). Nitrogen content was determined using colorimetric assays on a Technicon autoanalyzer (Bran and Lubbe Anal. Tech. 1987).

Instantaneous CERs were estimated on a dry mass basis according to von Caemmerer and Farquhar (1981). Daily CERs were calculated by integrating ten average CERs each during the diurnal 12 h and nocturnal 12 h. Daily CER on a whole-plant basis was calculated by integrating biomass and daily CER. Linear and nonlinear regressions were used to test for effects of time since the beginning of the experiment on CER per unit dry mass and per plant. Only leaf respiration of sugar maple in current CO_2 decreased with time (Reid 1990). Since no other parameters showed any significant time effect, data were averaged for further analysis. Analyses of variance were used to test for differences between species and CO_2 treatments on each of the gas exchange parameters using general linear models (SAS Institute 1986).

To compare the rate of biomass accumulation between CO_2 treatments, the simplest model of constant relative growth rate (RGR) was assumed (Landsberg 1986),

$$W_t = W_o e^{ct},$$

where W_t is plant dry mass at harvest time (g), W_o is initial dry mass (g), c is RGR ($days^{-1}$), and t is time interval (days). Because initial plant biomass could not be directly measured, it was estimated from initial stem length S_o according to the allometric relationship:

$$W_o = \alpha S_o^{\beta}$$

where α and β are constants. Taking logs and substituting for W_o yields the linear model:

$$\ln W_t = \ln\alpha + \beta \ \ln S_o + ct$$

Rate parameters c were compared using a Z-test (Kleinbaum and Kupper 1978). A similar model was used to compare the rate of change in root to shoot ratio (rt/sht).

Results

Diel gas exchange rates

Carbon exchange rates of both species were significantly affected by elevated CO_2, although their responses differed (Fig. 1). For beech, aboveground diurnal CERs increased 13% with CO_2 enrichment while sugar maple decreased 7%. Beech CERs remained lower than those of sugar maple. Decrease in diurnal CER of sugar maple was associated with a significant increase in the specific leaf mass under elevated CO_2 (data not shown). Aboveground nocturnal CERs were significantly decreased

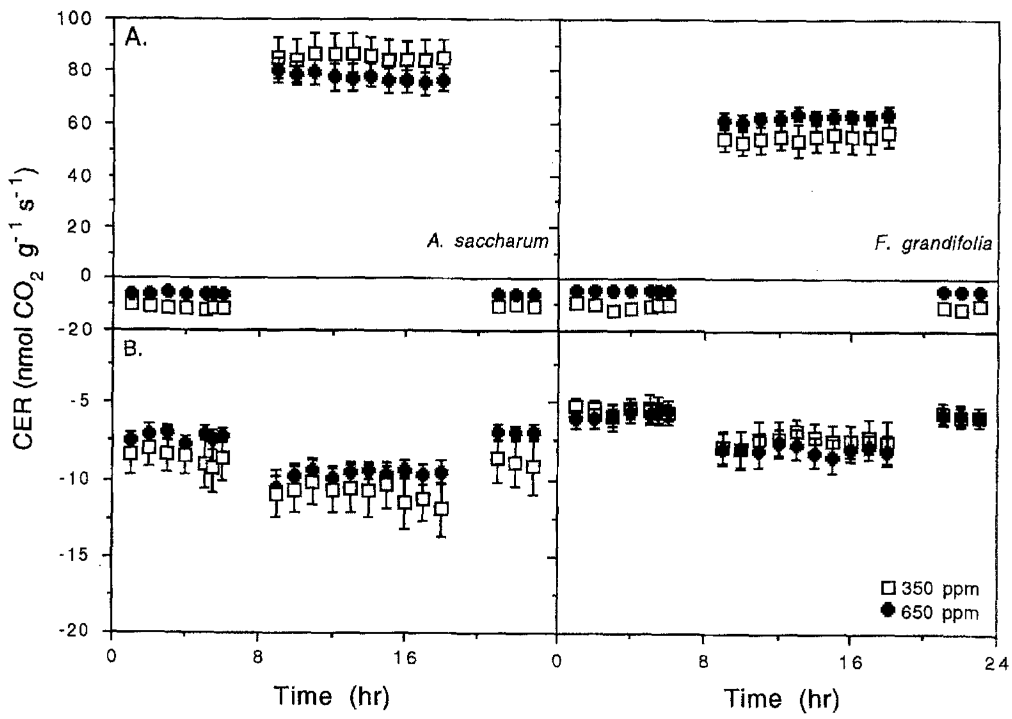

Fig. 1A, B Diel course of carbon exchange rates (CER) for seedlings of beech (*Fagus grandifolia*) and sugar maple (*Acer saccharum*) grown under different atmospheric CO_2. **A** Aboveground CER; **B** belowground CER. n=12, $\bar{x} \pm 1$ SEM with elevated CO_2 for both species, 50% for beech and 45% for sugar maple. The aboveground respiration rates of CO_2-enriched plants increased over the growing season whereas all other CER did not show any significant change with time (Reid 1990). Belowground CER was also significantly decreased by elevated CO_2 for sugar maple only. This decline was not sufficient to reach the low CERs of beech.

Whole-plant daily carbon budget

Whole-plant CERs showed a significant increase in daily carbon uptake with CO_2 enrichment for both species. The relative enhancement was greater for beech, resulting in similar daily carbon uptake for both species under CO_2 enrichment (Fig. 2A). The 62% increase in carbon uptake of beech was accompanied by a decrease in carbon lost to aboveground respiration (Fig. 2B). This was not the case for sugar maple because of no significant decline in respiration. Although the two species had similar aboveground respiration costs under contemporary CO_2, beech had less C lost under elevated CO_2. The proportion of daily carbon uptake lost to leaf respiration of beech declined under elevated CO_2, from 21% of C uptake at 350 $\mu l\ l^{-1}$ CO_2 to 7% at 650 $\mu l\ l^{-1}$ CO_2 (F=23.56, P=0.0001). For sugar maple, the proportion of carbon uptake lost to aboveground respiration was reduced from 13% at 350 $\mu l\ l^{-1}$ CO_2 to 7.5% at 650 $\mu l\ l^{-1}$ CO_2 (F=10.51, P=0.004). Carbon lost to maintenance respiration, which was similar for both species, was not affected by elevated CO_2 (Fig. 3A).

Total carbon lost to root respiration was not significantly affected by elevated CO_2 for either species although increased root respiration rates per plant were observed for sugar maple (Fig. 2C). Even though the two species had similar C lost to roots, the relative amount of carbon lost to roots went from 55% to 41% at 350 and 650 $\mu l\ l^{-1}$ respectively for beech, and 39% and 48% respectively for sugar maple. Root maintenance respiration was significantly affected by elevated CO_2 for both species but in different ways (Fig. 3B).

For beech, root maintenance respiration was enhanced under elevated CO_2 whereas, for sugar maple, the maintenance cost was decreased. Overall root respiration was the major carbon cost for seedlings of both species.

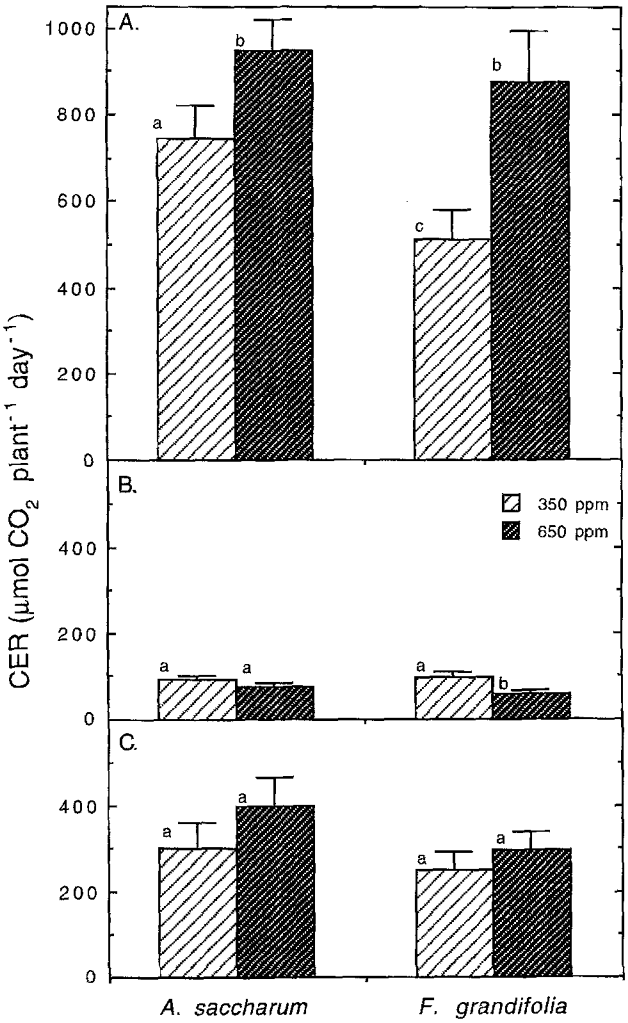

Fig. 2A–C Daily CER for beech and sugar maple grown under different atmospheric CO_2 concentrations and low irradiance. **A** carbon uptake; **B** aboveground carbon loss; **C** root carbon loss. Each histogram represents the average daily CER for the growing season; n=10–12, errors are ±1 SEM. (Different *letters* indicate significant difference at P≤0.05)

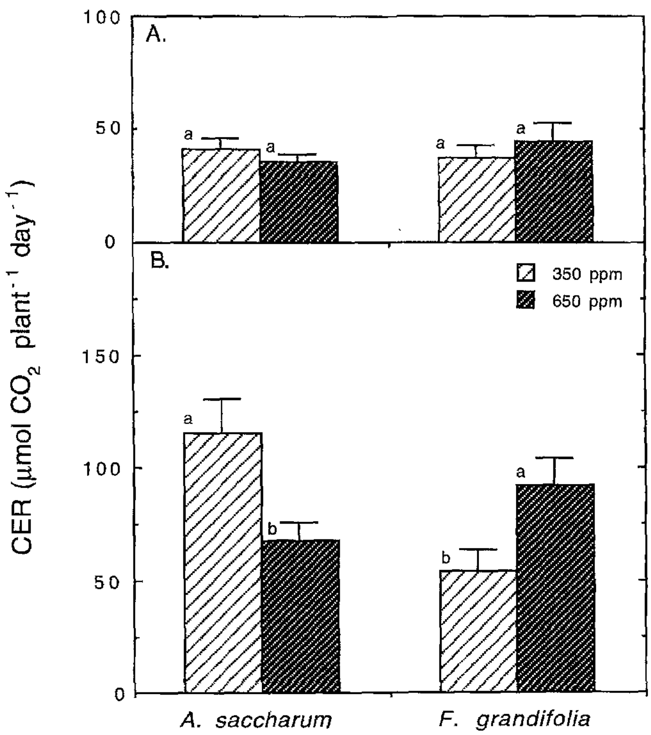

Fig. 3A, B Maintenance respiration for beech and sugar maple grown under different atmospheric CO_2 concentrations and low irradiance. Each histogram represents the average maintenance respiration for the growing season, as determined by the starvation method. **A** aboveground carbon loss; **B** belowground carbon loss. n=10–12, errors are ± 1 SEM (Different *letters* indicate significant difference at P≤0.05)

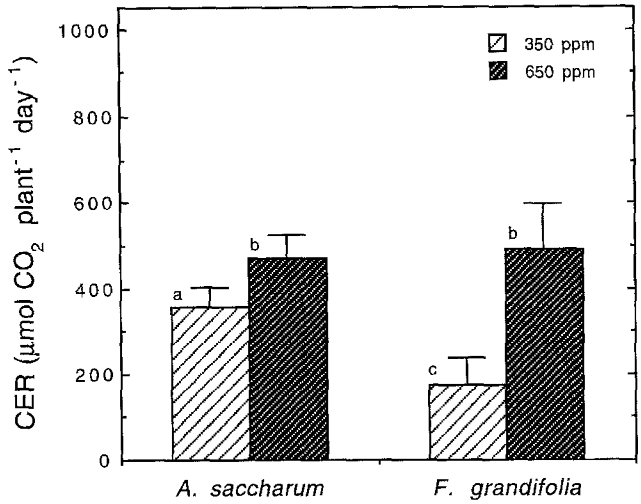

Fig. 4 Net daily carbon gain for beech and sugar maple grown under different atmospheric CO_2 and low irradiance. Each histogram represents the average net gain for the growing season calculated from daily CER. n=10–12, errors are ±1 SEM. (Different *letters* indicate significant difference at $P \leq 0.05$)

Net daily carbon gain was calculated from daily uptake and losses. Under contemporary CO_2, the carbon gain of beech was significantly lower than that of sugar maple (Fig. 4). The carbon gain increased with CO_2 enrichment for both species but the enhancement was greater for beech. This larger increase for beech resulted in similar net C gain for both species.

Nitrogen content and biomass

Carbon dioxide enrichment did not significantly affect the N concentration of leaf tissue for either species (Fig. 5A). Beech leaf N content remained lower than sugar maple. A significant increase in N concentration of fine roots was observed with CO_2 enrichment for beech only (Fig. 5B). The increase was not sufficient to reach N levels similar to sugar maple fine roots.

The log-linear relationships of biomass accumulation against time derived from our model were more robust for beech than for sugar maple (Fig. 6). Relative growth rates estimated from these relationships were not significantly different between CO_2 treatments for beech or sugar maple (Z=1.124, P=0.13 for beech; Z=0.059, P=0.48 for sugar maple). In addition, both species had similar RGR when grown under contemporary or elevated CO_2 (Z=0.318, P=0.374 at 350; Z=0.925, P=0.176 at 650 μl l^{-1}). Log-linear relationships of change in rt/sht against time were derived from a model similar to the biomass one (Fig. 7). The model results in a strong increasing relationship of rt/sht with time for beech under contemporary CO_2 only. The rate of increase of rt/sht was constant under elevated CO_2 (c=0.015 at 350 and c=0.003 at 650 μl l^{-1}; Z=2.072, P=0.019) although the relationship was weak when grown under CO_2 enrichment. For sugar maple, rates of change were not significantly different from zero and from each other (c=0.009 at 350 and c=0.007 at 650 μl l^{-1}; Z=0.274, P=0.39). Under elevated CO_2, both species have constant patterns of root and shoot allocation.

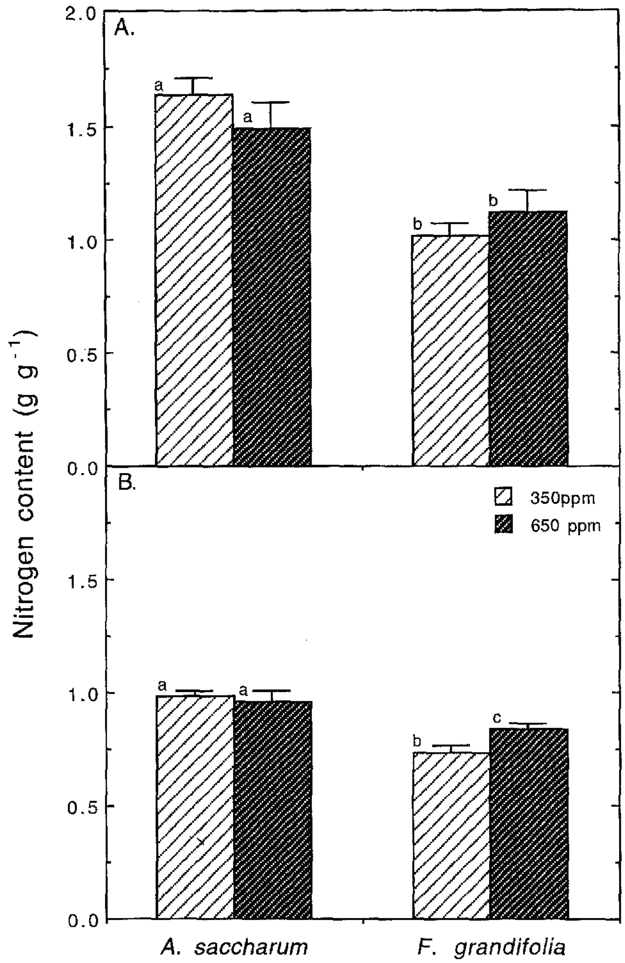

Fig. 5A, B Tissue nitrogen content for beech and sugar maple grown under different atmospheric CO_2 and low irradiance. **A** leaf and **B** small roots. Each *histogram* represents the average nitrogen concentration over the growing season. n=10–12, error bars are ±1 SEM (Different letters indicate significant difference at $P \leq 0.05$)

Discussion

In the understory, net carbon gain and survival of beech and sugar maple seedlings is by two basic patterns of CER (Reid 1990). The low photosynthetic capacity and respiration cost of beech is associated with slow continuous growth whereas the high photosynthetic capacity of sugar maple is associated with high maintenance cost of belowground biomass. These CER patterns and carbon balance are affected differently by elevated atmospheric CO_2, beech approaching a net balance comparable to sugar maple under CO_2 enrichment. The changes in CER, on a unit mass basis, are discussed first followed by the daily CER and resulting carbon balance.

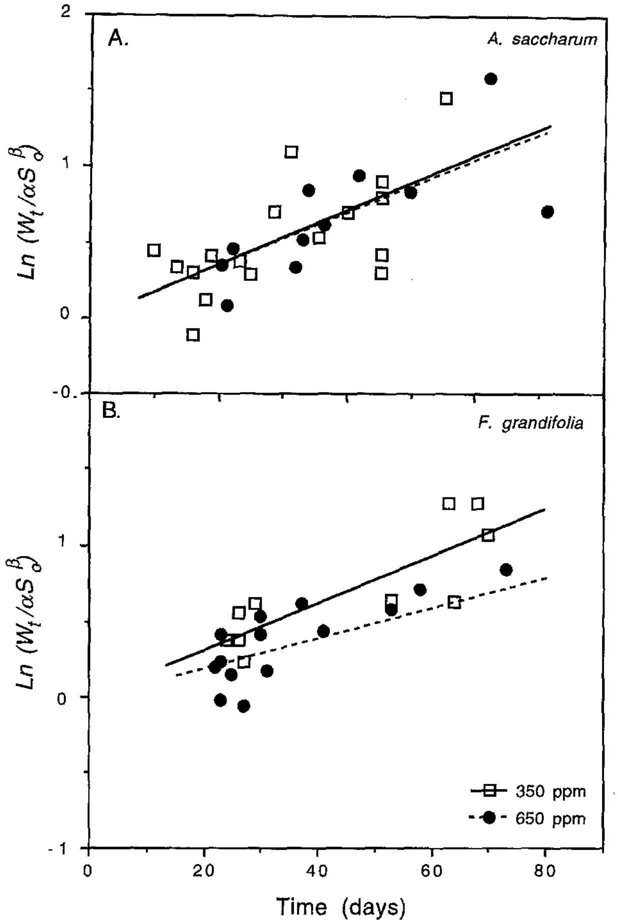

Fig. 6A, B Log-linear regression for total biomass accumulation of beech and sugar maple grown under different CO_2 concentrations. For beech, ln $(W_t/\alpha\ S_o^\beta)=0.015t$, $r^2=0.850$ at 350 µl l^{-1}, and ln $(W_t/\alpha\ S_o^\beta)=0.010t$, $r^2=0.855$ at 650 µl l^{-1} CO_2. Slopes are not significantly different between CO_2 treatments ($Z=1.124$, $P=0.131$). For sugar maple, ln $(W_t/\alpha\ S_o^\beta)=0.016t$, $r^2=0.461$ at 350 µl l^{-1}, and ln $(W_t/\alpha\ S_o^\beta)=0.015t$, $r^2=0.574$ at 650 µl l^{-1} and slopes are not significantly different between CO_2 treatments (Z=0.059, P=0.48)

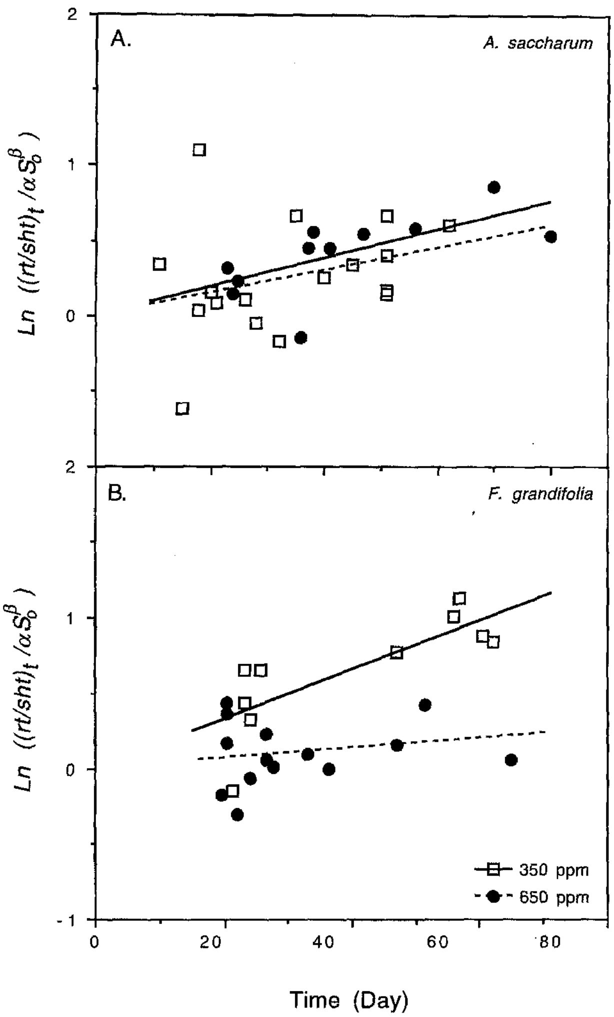

Fig. 7A, B Log-linear regression for increase in root to shoot ratio for beech and sugar maple grown under different CO_2 concentrations. For beech, ln $[(rt/sht)_t/\alpha\ S_o^\beta]=0.015t$, $r^2=0.623$ at 350 µl l^{-1}, and ln $[(rt/sht)_t/\alpha\ S_o^\beta]=0.003t$, $r^2=0.175$ at 650 µ l l^{-1} CO_2 and slopes are significantly different between CO_2 treatments ($Z=2.072$, $P=0.019$). For sugar maple, ln $[(rt/sht)_t/\alpha\ S_o^\beta]=0.007t$, $r^2=0.92$ at 350 µl l^{-1}, and ln $[(rt/sht)_t/\alpha\ S_o^\beta]=0.009t$, $r^2=0.410$ at 650 µl l^{-1} and slopes are not significantly different between CO_2 treatments ($Z=0.274$, $P=0.39$)

Diel carbon exchange rates

Constant diurnal or nocturnal aboveground CERs were observed for both species under both CO_2 treatments. A decline in diurnal photosynthetic rate is often associated with increasing accumulation of photosynthates in the leaf (Herold 1980). A faster decline usually occurs in plants grown in elevated CO_2 because of enhanced carbohydrate accumulation (Azcòn-Bieto 1983). The diurnal CERs measured here were low enough to prevent disruptive starch buildup in the chloroplast (Cave et al. 1981) and any inhibition of photosynthesis during the day even though elevated CO_2 compensated for the low irradiance and increased CER. Nocturnal leaf CER may also decrease through the night because of depletion of starch reserves for respiratory substrate (McCree and Silsbury 1978; McCree and Kresovich 1978). Since a greater carbohydrate pool may be available for respiration under elevated CO_2, a higher respiration rate may be maintained through the night. The constant nocturnal leaf CER measured here may reflect low diurnal activity rather than available carbohydrate pools. The lower nocturnal CER observed under CO_2 enrichment suggests reduced metabolic rate and substrate utilization. These low nocturnal CERs with elevated CO_2 may not deplete the starch pool overnight and, therefore, contribute to the gradual buildup of carbohydrates during the day. This buildup is suggested by an increase in specific leaf mass and leaf starch levels (Reid 1990). Reduced leaf nocturnal CERs were similar to the decreased respiration reported for wheat (Gifford et al. 1985).

The higher average photosynthetic rates observed in beech under elevated CO_2 are consistent with gas exchange rates observed in other studies with tree seedlings (Rogers et al. 1983; Tolley and Strain 1985; Williams et al. 1986; Hollinger 1987). This higher rate under low irradiance results from a decreased light compensation point (Tolley and Strain 1984). The lower average photosynthetic rate of sugar maple under elevated CO_2 is associated with a significant increase in specific leaf mass, although photosynthesis per leaf area is enhanced under elevated CO_2. Similar decreases of photosynthetic rates with CO_2 enrichment are reported for tropical tree seedlings (Oberbauer et al. 1985).

Species-specific root respiration rates presented here suggest two responses to elevated atmospheric CO_2: no change for beech and a decline for sugar maple. Roots grow in a CO_2-rich environment; thus root gas exchange is likely to be affected by an increase in atmospheric CO_2 only indirectly. The indirect effect is because of the close association of root and shoot activity (Szaniawski 1981; Szaniawski and Adams 1974). Such association is demonstrated by the constant nocturnal and diurnal root respiration rates of both species studied being well correlated with constant shoot activity for each CO_2 level. With elevated atmospheric CO_2 a greater translocation of photosynthates to the roots, because of enhanced photosynthesis, may increase substrate availability for root activity. Hence a greater respiration rate is expected. However, the lack of change or decrease in respiration shown by both species suggests other physiological responses to CO_2 enrichment. The constant root respiration rate of beech was accompanied by an increase in N content of fine roots, indicating either change in tissue composition, specific root length and/or reduced cost of N uptake. Nitrogen, which was readily available because plants were well-fertilized throughout the experiment, could be taken up at low cost to supply aboveground biomass. A more efficient utilization of energy may result in increased root activity rather than enhanced respiration rate. The decline in respiration of sugar maple was associated with a constant N content, again suggesting changes in biomass composition or lowered cost of acquiring N and nutrient use efficiency. Since energetics of ion transfer are an important component of root respiration (Veen 1980), the decrease in root respiration in elevated CO_2 may partly be due to a decrease in ion transfer cost.

Whole-plant daily carbon balance

Studies addressing whole-plant photosynthesis for young seedlings show a gradient from little to significant enhancement under CO_2 enrichment (e.g. Tolley and Strain 1985; Wong and Dunin 1987; Reekie and Bazzaz 1989). In the present study, whole-plant carbon uptake was increased with elevated CO_2 for seedlings of beech and sugar maple because of increased photosynthetic rate as well as leaf mass. Beech seedlings showed the greatest response, where gross productivity increased 62%. Productivity increased less than 26% for sugar maple, because the enhanced specific leaf mass with elevated CO_2 masked the increase observed on a leaf area basis. Carbon dioxide enhancement of daily carbon uptake for both species suggests that low-light stress is ameliorated under elevated CO_2, and this effect is more beneficial to beech than to sugar maple. The net benefit, however, is dependent on the respiratory costs associated with such increases.

Effects of elevated CO_2 on aboveground respiration range from decreased respiration for salt marshes (Drake 1992), to no change in respiration for *Plantago* (Poorter et al. 1988), to increased respiration of herbaceous perennial plants (Nijs et al. 1989a, b). The total carbon loss to respiration for beech aboveground biomass was reduced under elevated CO_2 despite greater leaf mass. For sugar maple, aboveground respiration remained constant because the reduced respiration rate was accompanied by increasing leaf mass with elevated CO_2. The total root respiratory cost of the seedlings did not change in elevated CO_2 because lower root respiration per unit mass was accompanied by an increase in root biomass. Roots as a carbon sink were not limiting because of optimum fertilization and large soil to root volume used in the experiment. Resulting carbon budgets differed for each species since the relative cost of supporting the beech root system was reduced whereas it increased for sugar maple.

Each component of respiration may be affected by elevated CO_2 although no satisfactory mechanisms are known (Amthor 1991; Ryan 1991). Maintenance respiration, which represents mostly protein turnover, is well correlated with tissue N content (Merino et al. 1982; Irving and Silsbury 1987). Since N content frequently decreases in CO_2-enriched plants (Norby et al. 1986), maintenance respiration may decline. For beech and sugar maple, the lack of change in aboveground maintenance respiration is concurrent with no change in N concentration. On the other hand, enhanced maintenance respiration of beech roots was correlated with increased N concentration in fine roots. Root maintenance respiration was significantly decreased for sugar maple even though N concentration remained constant. This suggests again that the cost of ion uptake and maintenance of the ion gradient may be reduced with elevated CO_2. With elevated CO_2, growth respiration may be increased because of a greater biomass production. However, change in tissue composition with elevated CO_2 may decrease construction cost. For example, non-structural carbohydrates, which increase under elevated CO_2 (Sasek et al. 1985), are less expensive to synthesize than proteins (Williams et al. 1987). Estimates of growth respiration, by subtracting maintenance from total respiration, do not change with elevated CO_2 for beech and sugar maple. This suggests lower construction cost because more biomass was produced.

The net carbon gain, which is derived from daily CO_2 fluxes, illustrated differential effects of elevated CO_2 on beech and sugar maple. For beech, the net carbon gain more than doubled under elevated CO_2 because the 62% increase in carbon uptake was accompanied by a 39% reduction in respiratory cost of the whole-seedling, thus ameliorating the C budget both in C uptake and losses. In contrast for sugar maple, the carbon balance was not changed because proportional carbon partitioning was maintained: the 26% increase in carbon input was accompanied by a 23% increase in root respiration. The stimulation of C uptake was thus counteracted by increased C losses. These data indicate a strong growth enhancement of beech with CO_2 enrichment. An index of carbon use efficiency (CUE), defined as the ratio of biomass accumulation to carbon uptake, was used to compare the efficiency of carbon utilization between the two species. This index is similar to biosynthetic efficiency but also integrates maintenance cost. Under current atmospheric conditions, CUE was 8% lower for sugar maple than beech. With CO_2 enrichment, CUE increased for both species and species difference increased to 14% because of a greater growth enhancement for beech (21%). This indicates a larger increase of beech productivity than sugar maple with CO_2 enrichment. Beech was also found to be the most responsive to elevated CO_2 among seven co-occuring northeastern trees (Bazzaz et al. 1990).

Growth models derived from biomass indicate little effect of CO_2 treatments on the relative rate of biomass accumulation. Lack of change in the rates of biomass accumulation and rt/sht with CO_2 treatments for sugar maple support the trends observed with the net carbon gain. Estimates of RGR increased 10–15% for beech in both CO_2 treatments. Similarity in RGR may be due to the increase in rate of change of rt/sht of beech during growth under current CO_2 conditions. More carbon is being invested in non-photosynthetic, respiring tissue under current conditions but not under elevated CO_2. The inherent slow growth of both species in low light used in this experiment may make detection of significant elevated CO_2 effects difficult in the short time scale used here. Effects may be visible on a time scale of years because of compounding C investment. Responses of seedlings of woody species to a doubling of atmospheric CO_2 range from no change in growth (Reekie and Bazzaz 1989; Norby and O'Neill 1989), through significant increases in biomass (Brown and Higginbotham 1986; Conroy et al. 1988; Hollinger 1987; Norby et al. 1987), to a doubling of biomass (Norby et al. 1986) depending on species and limiting environmental factors. Differences reported here for beech and sugar maple grown in similar low-light environments indicate that changes in net C gain by modified C losses as well as C uptake with CO_2 enrichment have a small impact on growth over one growing season.

Bazzaz et al. (1990) reported that, of seven tree species, the three shade-tolerant ones were the most responsive to CO_2 enrichment. Among shade-tolerant species, the trade-offs in morphological and physiological plasticity may be changed with CO_2 enrichment. The advantages conferred by the less physiologically active carbon budget of beech over sugar maple under low light are emphasized in CO_2-enriched atmospheres. Even though net carbon gain of seedlings of beech and sugar maple was enhanced with CO_2 enrichment, the pattern of C balance was altered more for beech, and the subsistence of beech seedlings in the low light of the understory may be extended.

Models of effects of climatic change during the Pleistocene on species distribution suggest that beech and sugar maple have migrated from the south to their current northern range limits (Davis and Zabinski 1992). Withdrawal of beech from its present southern range is predicted to be greater than sugar maple, because sugar maple will be favored over beech with global warming in deciduous forests. Other models which include decreases in soil moisture capacity also predict a greater enhancement of sugar maple than beech in northern hardwood forests (Pastor and Post 1988). Direct effects of CO_2 enrichment presented here, however, suggest that beech overall will benefit more from CO_2 fertilization than sugar maple, thus increasing competitive pressure on sugar maple. As pointed out by Bazzaz et al. (1990) in consideration of the probable increases in soil and air temperatures, integration of direct and indirect effects of elevated CO_2 on tree species must be examined in order to predict future species distribution and forest composition.

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